

Edge Computing in Industrial Communication

TEAM-A5

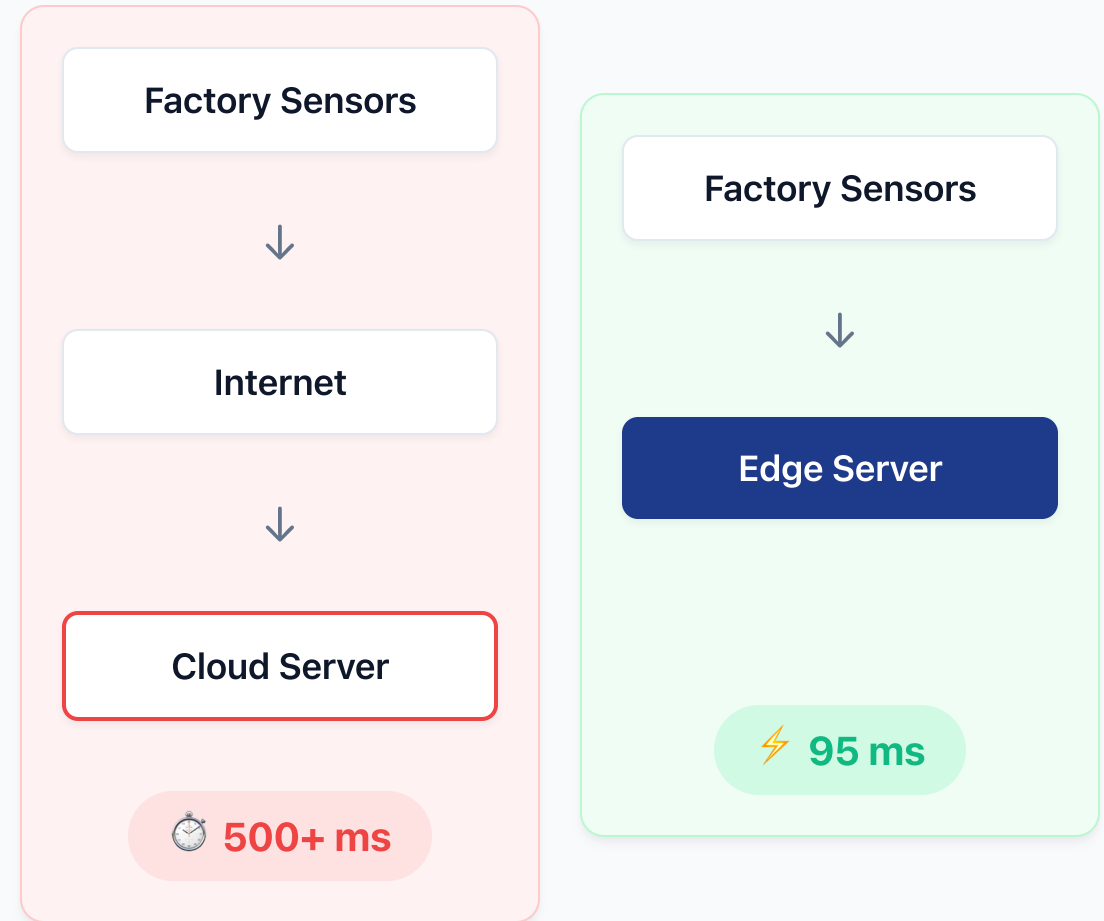


Why Do We Need Edge Computing?

The Problem:

- Billions of IoT devices generate data.
- Industrial systems need millisecond responses.
- Sending everything to the cloud causes delay.
- Delays can damage equipment and reduce safety.

Edge computing brings computation closer to the machines.



The Industrial Problem

Example: Robotic Welding Station

Cloud Processing

Weld defect detected



Data sent to Cloud



Response arrives



Machine Damaged (Too Late)

Edge Processing

Weld defect detected



Local Edge Node



Immediate Shutdown

How Edge Computing Works



Cloud Layer

Long-term storage & Historical Analysis

↑ Only important results go to cloud ↑



Edge Layer

Local processing, immediate actions

↑ Machines generate raw data ↑

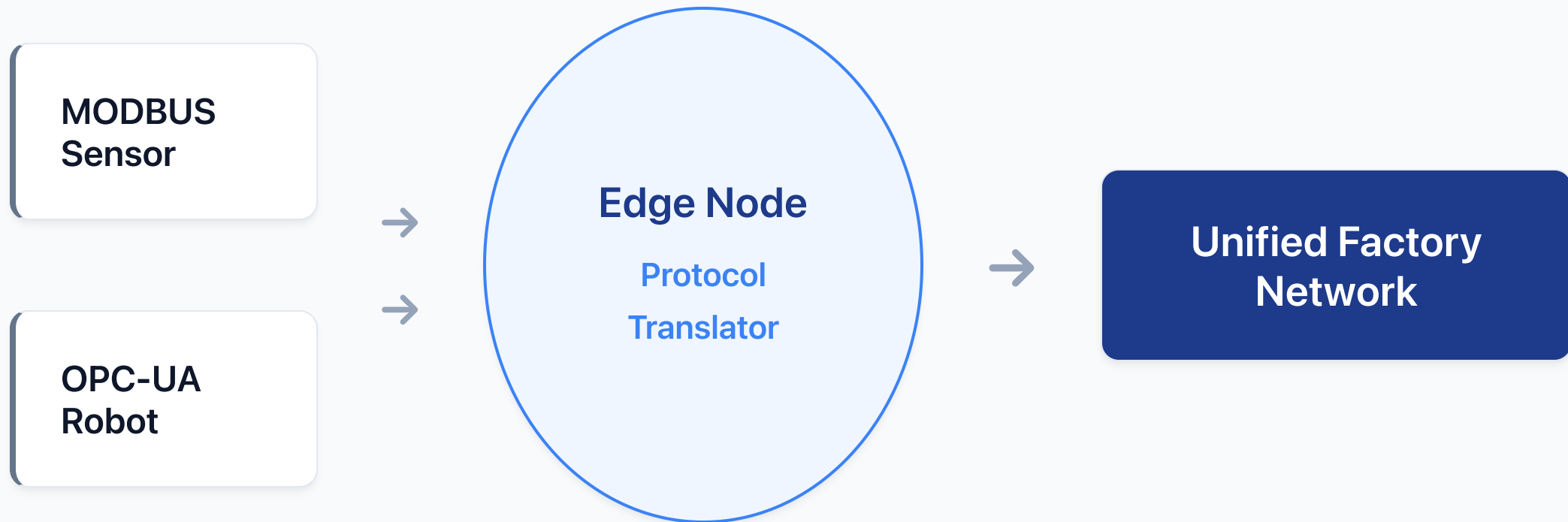


Industrial Devices

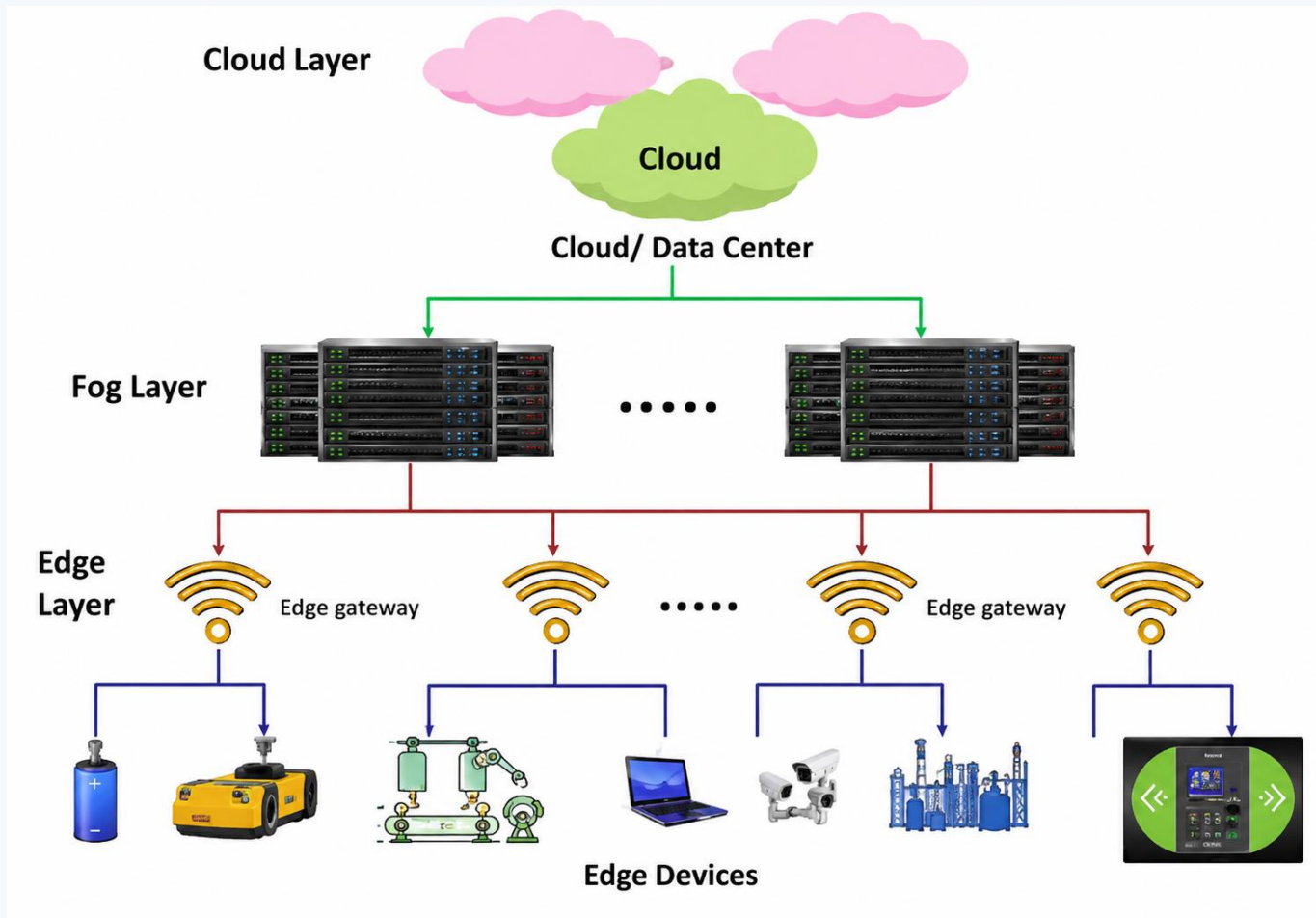
Sensors, PLCs, Cameras, Robotic Arms

Edge Node = Factory Translator

Different languages → Edge translates → Everyone communicates



Architecture of Edge Computing



Cloud layer

Long-term storage, dashboards, model training, ERP/MES reporting

Fog layer

Buffers and aggregates data between edge and cloud

Edge layer

Local processing, AI/fault detection, protocol translation

Edge devices

Sensors, cameras, PLCs, robots and IoT modules

Cloud vs. Edge Computing

Cloud Computing

Edge Computing

Processing Location	Central cloud servers	Near the device/user
Latency	Higher	Lower
Bandwidth Use	High — all data to cloud	Low — only summaries sent
Storage Capacity	Very high (unlimited)	Limited (local storage)
Reliability	Requires internet connection	Works offline
Best Use Case	Big data · ML training · ERP	Real-time control · Safety

Latency test Setup

Latency Definition

Tested in two environments: Paderborn home Wi-Fi (stable, private) · HSHL central Wi-Fi (shared, congested)

Client device

Mobile phone

Generated sensor-like data

Edge server

Ubuntu laptop

Processed the request locally

Cloud target

Internet server

External online server

Measured latency



Request sent



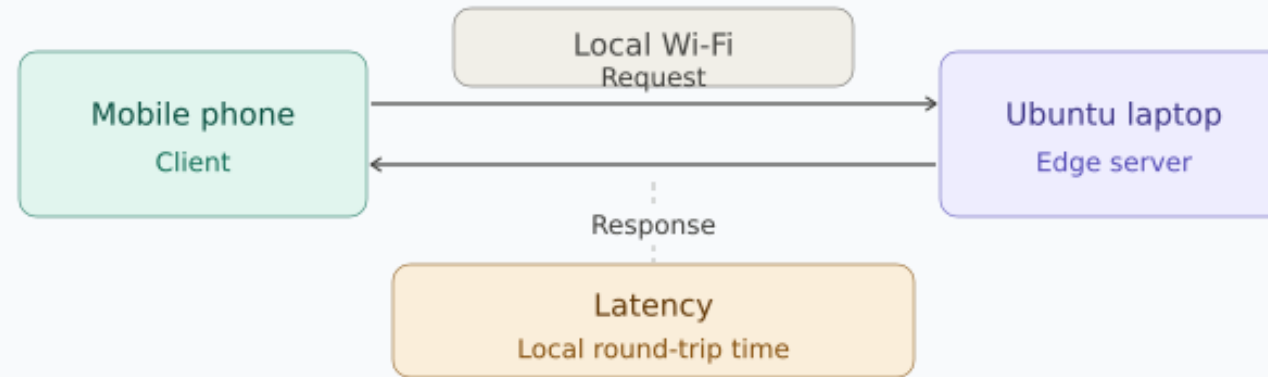
Response received

Time from sending a request until the response is received.

Only “HSHL” Wifi USER



Edge Latency Test



Test Procedure

Server: Ubuntu 22.04 running Python Flask

Client:

Mobile phone → `http://192.168.178.78:5000`

Terminal log:

```
POST /process HTTP/1.1" 200
192.168.178.117 → .78:5000
```

Status 200 OK confirms successful edge communication.

Stable local network

95 ms

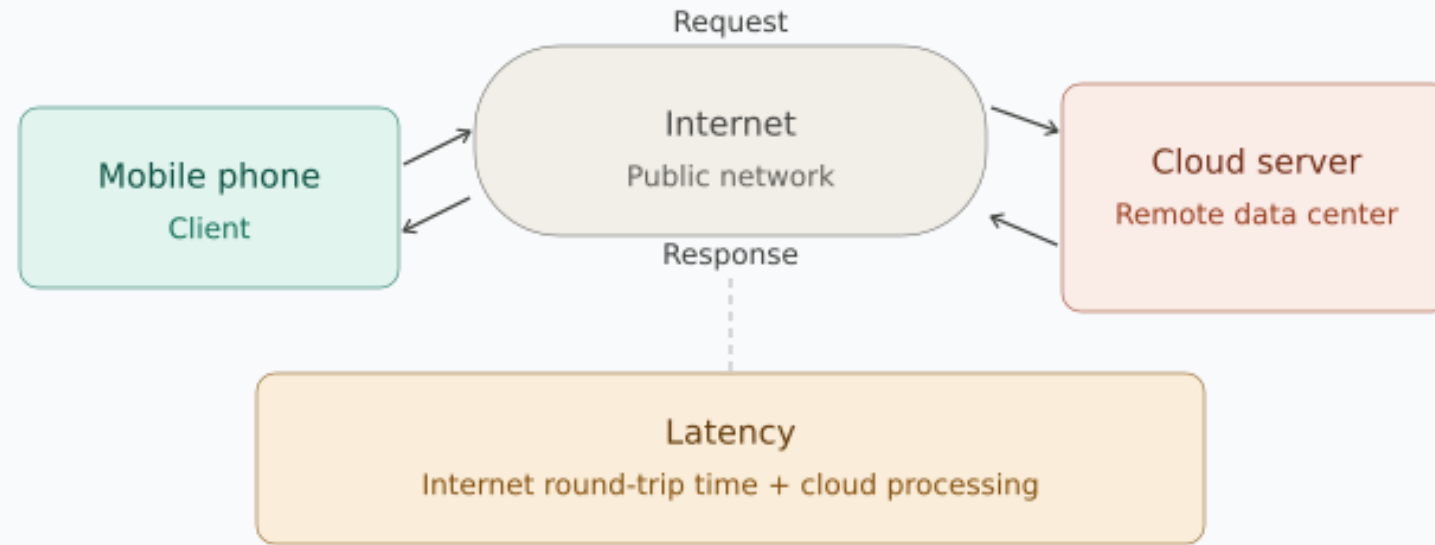
Paderborn home Wi-Fi

Shared network

422 ms

HSHL central Wi-Fi

Cloud Latency Test



Test Procedure

Method: HTTP request via public internet

Target:

`https://httpbin.org/get`

Command:

```
curl -w "%{time_total}s"
https://httpbin.org/get
```

Result: 0.510079 s = **510 ms** internet round-trip.

Measured cloud latency

510 ms

Paderborn home Wi-Fi

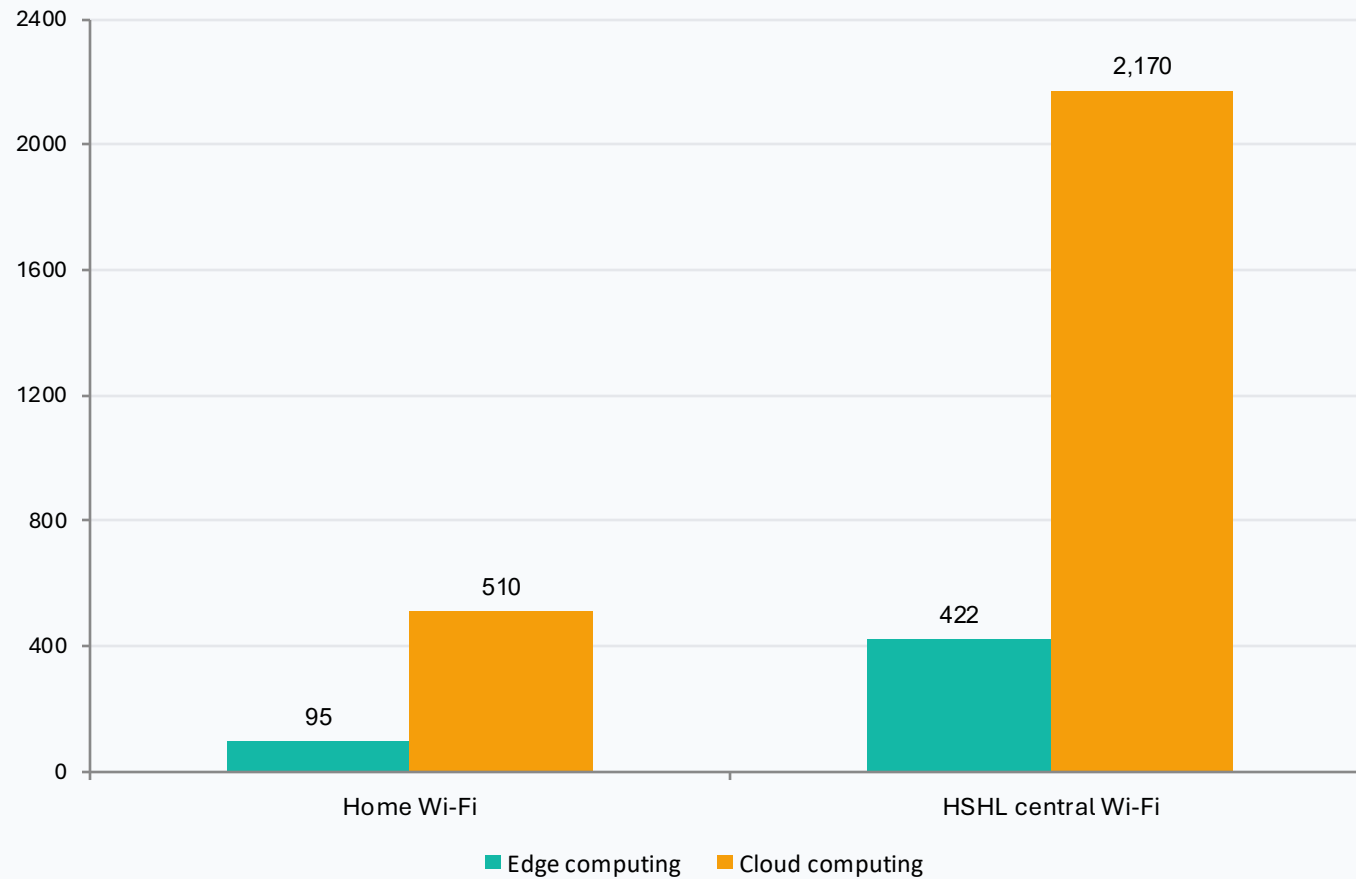
Worst shared network result

2170 ms

HSHL central Wi-Fi

Measured Results: Edge vs Cloud

Latency in milliseconds (lower is better)



Home Wi-Fi

95 vs 510 ms

Cloud took about 5.4× longer

HSHL Wi-Fi

422 vs 2170 ms

Cloud took about 5.1× longer

Latency Formulation

$$C = X / L + D / R$$

C = latency

X = distance • L = bandwidth • D = data rate • R = computing resources

Edge path



short X

Cloud path



large internet X

What increases delay?

- Edge reduces the distance X because processing is local.
- A shared network lowers available bandwidth L.
- Cloud routes are longer, so strict millisecond response is harder.

Edge vs. Cloud: Full Comparison

Criterion	Edge Computing	Cloud Computing
Latency (stable network)	95 ms	510 ms
Latency (shared network)	422 ms	2,170 ms
Speed advantage	5.1× – 5.4× faster in every test	
Offline operation	✓ Yes	✗ No
Suitable for real-time	✓ Yes	Limited
Network dependency	Local network	Internet required
Storage & analysis scale	Limited	✓ High

Edge is best for fast local actions. Cloud is best for heavy storage, training, reporting, and long-term analytics.

Key Benefits



Low Latency

Processing near the source cuts response times to milliseconds — essential for autonomous vehicles, robotics, and emergency healthcare.



Reduced Bandwidth

Only key results (alarms, status updates) travel to the cloud. Raw data stays local, saving network capacity.



Improved Privacy

Sensitive data (e.g. patient records) stays on-premise. Only necessary summaries reach the cloud.



Better Reliability

Edge systems continue operating when cloud connectivity is poor or lost — critical for remote factories.

From Experiment to Industrial System

Experiment	Industrial Equivalent
Mobile Phone	Sensor / PLC / Robot
Wi-Fi	Industrial Ethernet / OPC UA / 5G
Ubuntu Laptop	Edge Gateway
Flask Application	Edge Analytics
Latency Measurement	Response Time

Real Industrial Applications



PCB Assembly

Cameras inspect boards instantly. Detects missing components and rejects boards before moving down the line.



CNC Machining

Monitors vibration and motor current. Detects tool wear locally to prevent damage to expensive workpieces.



Robotic Welding

Evaluates voltage and weld times in milliseconds. Stops defective welds immediately.



Pharma Packaging

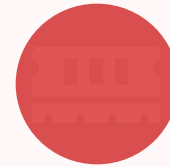
Verifies fill levels and label positions. Removes faulty or underfilled products directly on the spot.

Challenges & Limitations



Security Risks

Edge devices are distributed across many sites, making centralised protection difficult. A compromised device can impact the whole system.



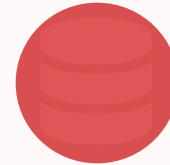
Limited Resources

Edge hardware has less compute, memory, and storage than cloud servers. Applications must be carefully optimised.



Management Complexity

Coordinating software updates, monitoring, and fault detection across hundreds of edge nodes is a significant operational challenge.



Data Consistency

Processing data at multiple sites makes it harder to maintain a single consistent view between edge and cloud systems.

Conclusion

- Edge computing is a key technology for modern distributed systems — not a replacement for the cloud, but a complement.
- The edge handles latency-sensitive local tasks (fault detection, safety response, quality control).
- The cloud handles large-scale storage, model training, and long-term analytics.
- Experimental results confirm edge processing is 5.1–5.4× faster across all network conditions tested.
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Thank you · Questions welcome